Final Exam Questions

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ADC: Sazzadul Islam
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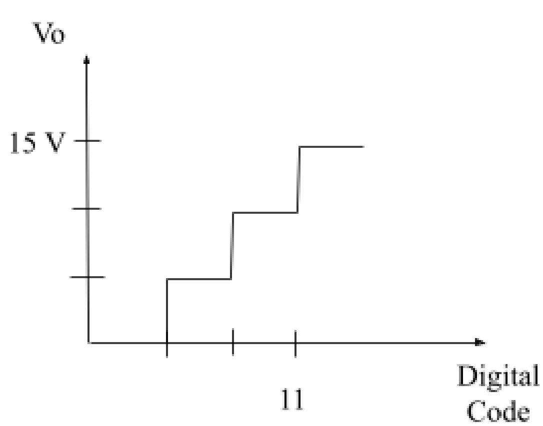
Please submit your questions by 22nd September 2024. Please make sure you make Set A and Set B of your question. Please provide a handwritten/latex solution with the marks distribution to your question.

- While posting your question here, please follow this format:

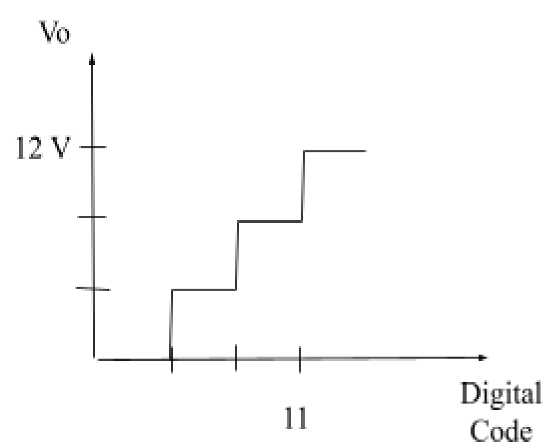
(a)	CO	Question	[Marks]
(b)	CO	Question	[Marks]
(c)	CO	Question	[Marks]

DAC [TNMF]

CO3	(a)	Identify the converter type and bits from the data given in Figure 1 transfer curve (Vo vs Digital Code). **Calculate** Vref when RF= 2R.	**[2+4]**
	(b)	Draw the converter with appropriate values from (a).	**[4]**
	(c)	**Calculate** all the outputs, Vo for the converter. Draw and fill the transfer curve of (a) with all the input and output values of the converter.	**[8+2]**



SET A



SET B

Solution

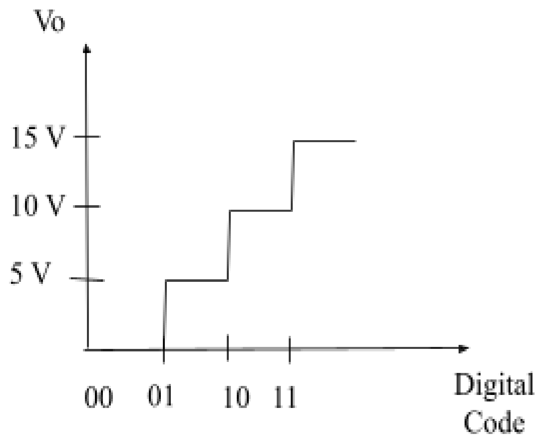
a) **Binary weighted DAC. 2 bit [2]**

Vo= -Vref*(2R/R)*[b1+b0/2] => Vref= -Vo/3= -15/3= -5 V [4] [SET A]
=> Vref= -Vo/3= -12/3= -4 V [4] [SET B]

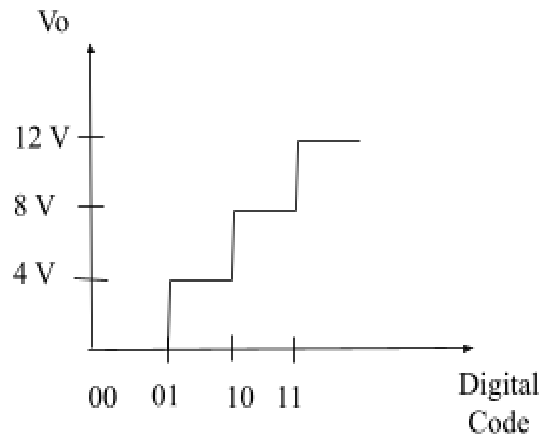
b) **2 bit DAC where RF=2R [4]**

c) **[8+2]**

Vin	Vo [SET A]	Vo [SET B]
00	0 V	0 V
01	5V	4V
10	10V	8V
11	15V	12V



SET A



SET B

Square Wave + CMOS [NMS]

Question (Set A) **[20]**

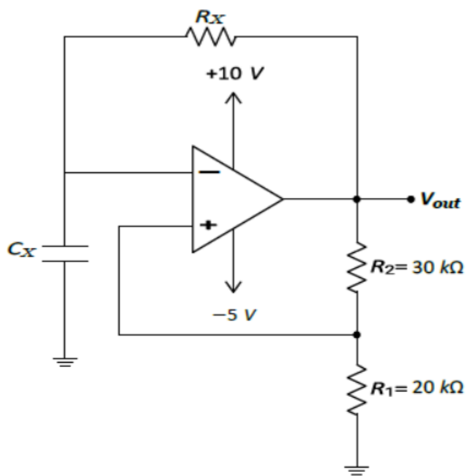


Figure 1

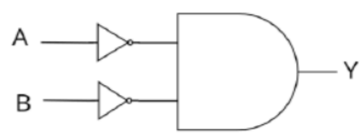


Figure 2

CO3	(a)	For the square wave generator in **figure 1, determine** V_{TH} and V_{TL} values of the Schmitt Trigger. *(Upper and lower threshold values)*	[3+3]
CO3	(b)	**Determine** the ***period***, ***time constant*** and ***duty cycle*** of the square wave generator circuit, given that the ***frequency*** is 100 Hz	[6]
CO1	(c)	Design an equivalent CMOS logic circuit for **figure 2** using the minimum number of NMOS and PMOS transistors. Also, provide the truth table showing the 'ON' and 'OFF' states of each MOSFET for every input test case.	[8]

Question (Set B) **[20]**

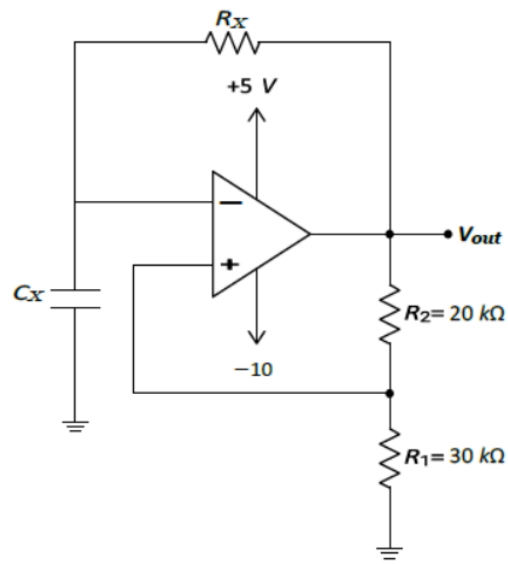


Figure 1

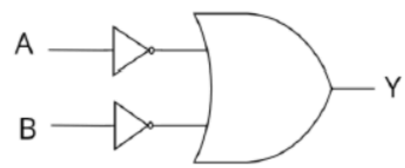


Figure 2

CO3	(a)	For the square wave generator in **figure 1, determine** V_{TH} and V_{TL} values of the Schmitt Trigger. *(Upper and lower threshold values)*	[3+3]
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Solutions:

Question (Set A) **[20]**

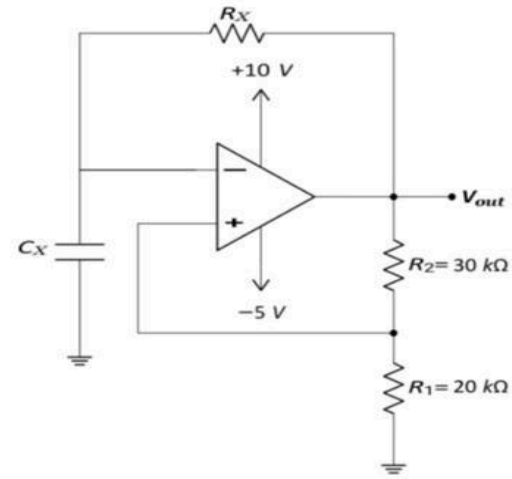

Figure 1

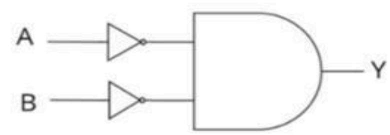

Figure 2

CO3	(a)	For the square wave generator in **figure 1, determine** V_{TH} and V_{TL} values of the Schmitt Trigger. *(Upper and lower threshold values)*	[3+3]
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Soln:

(a) Upper Threshold, $V_{TH} = \frac{R_1}{R_1+R_2} V_H$

$$= \frac{20K}{20K+30K} \times 10V$$

$$\boxed{V_{TH} = 4V}$$

Lower Threshold, $V_{TL} = \frac{R_1}{R_1+R_2} V_L$

$$= \frac{20K}{20K+30K} \times (-5V)$$

$$\boxed{V_{TL} = -2V}$$

(b) given, $f = 100\,Hz$

$\therefore$ period, $T = \frac{1}{f} = \frac{1}{100} = 10\,ms.$

$\rightarrow T_1 = R_x C_x \ln\left|\frac{V_H - V_{TL}}{V_H - V_{TH}}\right|$
$= R_x C_x \ln\left|\frac{10+2}{10-4}\right|$
$= R_x C_x \ln 2$

$\rightarrow T_2 = R_x C_x \ln\left|\frac{V_L - V_{TH}}{V_L - V_{TL}}\right|$
$= R_x C_x \ln\left|\frac{-5-4}{-5+2}\right|$
$= R_x C_x \ln 3$

$\therefore T = T_1 + T_2$

$\Rightarrow 10\,ms = R_x C_x (\ln 2 + \ln 3)$

$\Rightarrow R_x C_x = \frac{10\,ms}{\ln 2 + \ln 3} = 5.58\,ms$

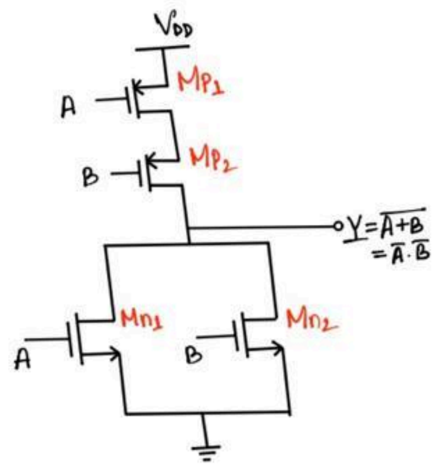
$\therefore T_1 = 5.58 \ln 2$
$= 3.86\,ms$

$\therefore$ Time constant, $\tau = R_x C_x = 5.58\,ms$

Duty cycle, $D = \frac{T_1}{T} \times 100\%$
$= \frac{3.86}{10} \times 100\%$
$= 38.6\%$ (Ans:)

(C)

given, $Y = \bar{A} \cdot \bar{B} = \overline{A+B}$



A	B	Y	M_{n1}	M_{n2}	M_{P1}	M_{P2}
0	0	1	OFF	OFF	ON	ON
0	1	0	OFF	ON	ON	OFF
1	0	0	ON	OFF	OFF	ON
1	1	0	ON	ON	OFF	OFF

Question (Set B) **[20]**

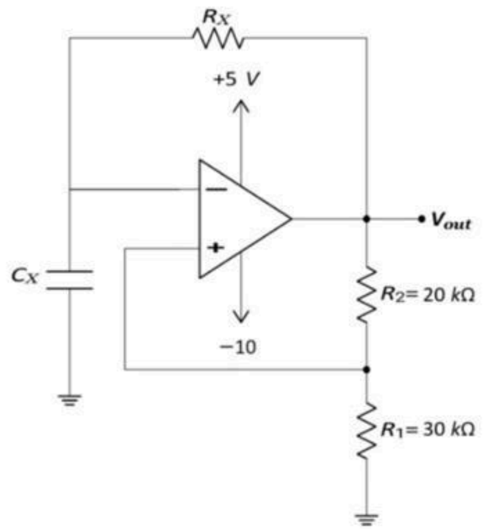


Figure 1

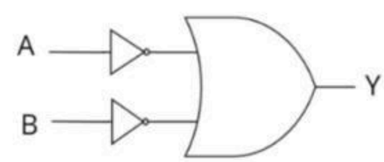


Figure 2

CO3	(a)	For the square wave generator in **figure 1, determine** V_{TH} and V_{TL} values of the Schmitt Trigger. *(Upper and lower threshold values)*	[3+3]
CO3	(b)	**Determine** the ***period, time constant*** and ***duty cycle*** of the square wave generator circuit, given that the ***frequency*** is 100 Hz	[6]
CO1	(c)	Design an equivalent CMOS logic circuit for **figure 2** using the minimum number of NMOS and PMOS transistors. Also, provide the truth table showing the 'ON' and 'OFF' states of each MOSFET for every input test case.	[8]

Soln:

(a) Upper threshold, $V_{TH} = \frac{R_1}{R_1+R_2} V_H = \frac{30K}{30K+20K} \times 5\,V$

$$\boxed{V_{TH} = 3V}$$

Lower threshold, $V_{TL} = \frac{R_1}{R_1+R_2} V_L = \frac{30K}{30K+20K} \times (-10V)$

$$\boxed{V_{TL} = -6\,V}$$

(b)

given, $f = 100\,Hz$

$\therefore$ period, $T = \frac{1}{f} = \frac{1}{100} = 10\,ms.$

$\rightarrow T_1 = R_x C_x \ln\left|\frac{V_H - V_{TL}}{V_H - V_{TH}}\right|$
$= R_x C_x \ln\left|\frac{5+6}{5-3}\right|$
$= R_x C_x \ln(5.5)$

$\rightarrow T_2 = R_x C_x \ln\left|\frac{V_L - V_{TH}}{V_L - V_{TL}}\right|$
$= R_x C_x \ln\left|\frac{-10-3}{-10+6}\right|$
$= R_x C_x \ln(3.25)$

$\therefore T = T_1 + T_2$

$\Rightarrow 10\,ms = R_x C_x (\ln 5.5 + \ln 3.25)$

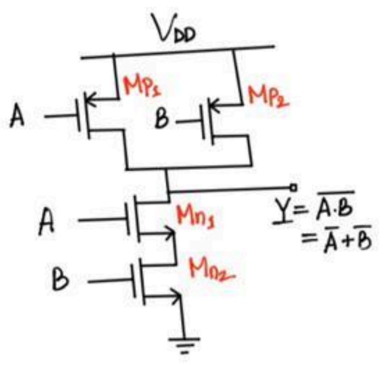
$\Rightarrow R_x C_x = \frac{10\,ms}{\ln 5.5 + \ln 3.25} = 3.46\,ms$

$\therefore T_1 = 3.46 \ln(5.5)$
$= 5.91\,ms$

$\therefore$ Time constant, $\tau = R_x C_x = 3.46\,ms$

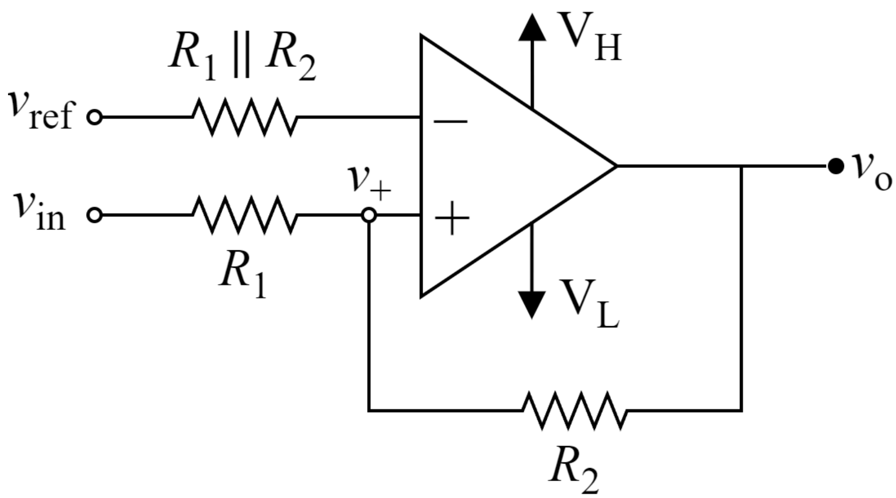
Duty cycle, $D = \frac{T_1}{T} \times 100\%$
$= \frac{5.91}{10} \times 100\%$
$= 59.1\%$ (Ans:)

(c) $Y = \bar{A} + \bar{B} = \overline{A \cdot B}$



A	B	Y	M_{n1}	M_{n2}	M_{P1}	M_{P2}
0	0	1	OFF	OFF	ON	ON
0	1	1	OFF	ON	ON	OFF
1	0	1	ON	OFF	OFF	ON
1	1	0	ON	ON	OFF	OFF

Schmitt Trigger [QSH]



Given,

For Set A : **V_H = 10V** , **V_L = -10V** , **v_{ref} = 2V**
For Set B : **V_H = 9V** , **V_L = -9V** , **v_{ref} = 3V**

(a)	CO3	What **type** of schmitt trigger circuit is this **(inverting or non-inverting)**?	[2]
(b)	CO3	**Draw** the **VTC (v_{in} vs v_o)** of this schmitt trigger circuit. Correctly **label** the plot (Depict **V_{TH}, V_{TL}, V_H, V_L, Hysteresis Width**). Given, **For Set A** : **R_1 = 1kΩ and R_2 = 2kΩ** for question (b) only. **For Set B** : **R_1 = 1kΩ and R_2 = 3kΩ** for question (b) only.	[8]
(c)	CO3	**Determine** the value of R_1 and R_2. Given, **For Set A** : **$R_1 \parallel R_2$ = 4kΩ; V_{TH} = 5V** and **V_{TL} = 0V** is required, for question (c) only. **For Set B** : **$R_1 \parallel R_2$ = 6kΩ; V_{TH} = 9V** and **V_{TL} = 0V** is required, for question (c) only.	[10]

Solution:

Set A

(a) Non-inverting

(b) $V_{TH} = v_{ref}\left(\frac{R_1+R_2}{R_2}\right) + \left(-\frac{R_1}{R_2}\right)V_L$

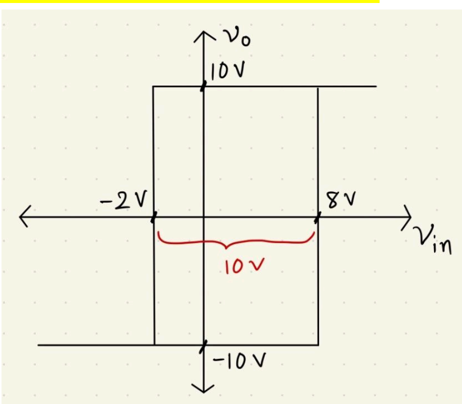
$V_{TH} = 8\ V$

$V_{TL} = v_{ref}\left(\frac{R_1+R_2}{R_2}\right) + \left(-\frac{R_1}{R_2}\right)V_H$

$V_{TL} = -2\ V$

$Hysteresis\ Width = V_{TH} - V_{TL}$

$Hysteresis\ Width = 10\ V$



(c) $Hysteresis\ Width = \left(-\frac{R_1}{R_2}\right)\left(V_L - V_H\right)$

$\Rightarrow \frac{R_1}{R_2} = \frac{1}{4}$

$\Rightarrow R_2 = 4R_1$

Given, $R_1 || R_2 = 4k\Omega$

$\Rightarrow R_1 = 5k\Omega$

$\Rightarrow R_2 = 20k\Omega$

Set B

(a) Non-inverting

(b) $V_{TH} = v_{ref}\left(\frac{R_1+R_2}{R_2}\right) + \left(-\frac{R_1}{R_2}\right)V_L$

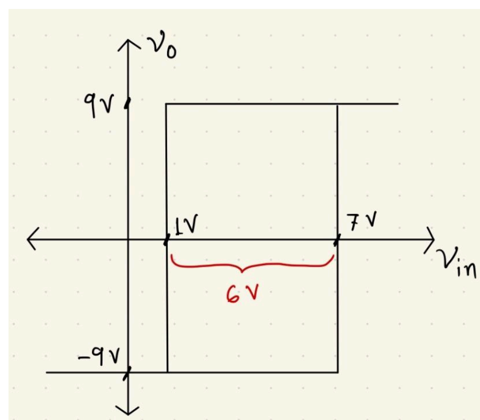
$V_{TH} = 7\ V$

$V_{TL} = v_{ref}\left(\frac{R_1+R_2}{R_2}\right) + \left(-\frac{R_1}{R_2}\right)V_H$

$V_{TL} = 1\ V$

$Hysteresis\ Width = V_{TH} - V_{TL}$

$Hysteresis\ Width = 6\ V$



(c) $Hysteresis\ Width = \left(-\frac{R_1}{R_2}\right)\left(V_L - V_H\right)$

$\Rightarrow \frac{R_1}{R_2} = \frac{1}{2}$

$\Rightarrow R_2 = 2R_1$

Given, $R_1 || R_2 = 6k\Omega$

$\Rightarrow R_1 = 9k\Omega$

$\Rightarrow R_2 = 18k\Omega$

ADC [SZZ]

Solution:

(a)

Set A & Set B

Quantization Range (V)	Digital Output	LED 1	LED 0
0 - 1.25	00	OFF	OFF
1.25 - 3.75	01	OFF	ON
3.75 - 6.25	10	ON	OFF
6.25 - 10	11	ON	ON

(b)

Set A

Time (s)	Sampled value (V)	Quantization Range (V)	Digital Output
0.0	5.00	3.75 - 6.25	10
0.5	7.40	6.25 - 10	11
3.0	5.71	3.75 - 6.25	10
4.0	1.22	0 - 1.25	00

Set B

Time (s)	Sampled value (V)	Quantization Range (V)	Digital Output
4.0	1.22	0 - 1.25	00
3.0	5.71	3.75 - 6.25	10
0.5	7.40	6.25 - 10	11
0.0	5.00	3.75 - 6.25	10